

ISAIAH KOL

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SUMMARY

Mechanical engineering student with a background in competitive robotics and experience in lab leadership, designing and building mechanical systems through CAD, fabrication, and iterative prototyping.

EDUCATION

University of Vermont | Burlington, VT Aug 2024 – Present
Bachelor of Science, Mechanical Engineering | **Expected May 2028**
- GPA: 3.80 | Dean's List

EXPERIENCE

UVM FabLab | Burlington, VT Dec 2025 – Present
Operations Manager
- Improve workflow, equipment use, and space organization to optimize lab efficiency and engagement.
- Coordinate technicians and oversee day-to-day lab and equipment operations.
- Lead lab operations while continuing hands-on technician work.
Technician Sep 2025 – Dec 2025
- Operated 15+ 3D printers and other equipment for student projects and university research.
- Assisted students and researchers with CAD, project development, and equipment use.
- Ensured lab organization, safety standards, and routine machine maintenance.

Dartmouth College Dining Services Line Cook & Counterworker | Hanover, NH Aug 2022 – Aug 2024
- Balanced customer service with cooking and food preparation in a fast-paced dining environment.

Gap-Year: VEX Robotics Team Coach | Thetford, VT Aug 2023 – Jun 2024
- Mentored students ages 12-18 in CAD, design process, safe fabrication, and competition strategy.

VEX Robotics Team 4886-B | Various Locations, USA Sep 2017 – Jun 2023
- Shared responsibility for robot design, fabrication, controls, and documentation on a two-person team.
- Four-time VEX World Championship competitor; 2023 NH/VT State Champion; recipient of three Design Awards and three Excellence Awards at the regional (NH/VT) level.

Internship at Hypertherm & Fujifilm Dimatix | Lebanon, NH Aug 2021 – Jan 2022
- Built foundational skills in CAD and CNC/manual machining through guided projects.
- Learned manufacturing practices and engineering processes in professional environments.

PROJECTS

Cycloidal Drive
- Designed and fabricated a 22:1 cycloidal gearbox for a NEMA 17 stepper motor.
- Iterated geometry and FDM print tolerances to minimize backlash.
- Achieved highly compact form by matching axial gearbox length to motor shaft length (20 mm).
- Tested actuation with an Arduino Uno as part of an in-progress robot arm.

Universal Spool Adapter for Filament Recycler
- Identified a geometry constraint in FabLab's PLA recycling system.
- Designed and 3D printed a fixture that enabled use of generic spools on the machine.
- Reduced empty spool waste and eliminated the need for proprietary spools.

SKILLS

Design: SOLIDWORKS, AutoCAD, Design for Manufacturability (DFM), Technical Documentation
Fabrication: FDM 3D Printing, Basic Machining, Automotive Repair
Software: Excel, PowerPoint, Basic Python, FDM Slicing Software